

Guidelines for making an *open, documented and sustainable* 3D model of a complex mechanism

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In this note, I want to share my experience of creating 3D models of mechanisms, and specifically of a number of issues that I consider important for the sustainability of 3D models.

1 The context: scientific heritage

As an independent researcher, I have been involved since the beginning of this century in the study and documentation of scientific instruments. In particular, I have spent a great deal of time studying complex astronomical clocks, such as the Strasbourg astronomical clock. Such clocks not only contain many parts, gears, and so on, but also parts that move intermittently. Modelling these mechanisms in 3D is much more complex than modelling rigid assemblies, such as architectural structures.

Back in 2001, I began creating a model of an emblematic part of the Strasbourg clock, namely the mechanism that computes the date of Easter. This was programmed in C using OpenGL; the teeth of the gears were approximated by trapezoidal shapes, and I obtained a rough rendering of this complex mechanism. During the following years, I worked on other mechanisms and developed new techniques. At the same time, I was also deeply involved in the survey of turret clocks, and my work always focused on documentation.

In 2014, I completely redesigned the construction of gears and had a couple of gears printed in 3D. Little by little, I continued to develop

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new tools, and eventually, in 2019, I was able to create a 3D model of the lunar panel of Rittenhouse's orrery in Philadelphia. This, in turn, made it easier to create the 3D model of the destroyed clock of Paris Notre-Dame Cathedral in 2020. In 2025–2026, I made a 3D model of the planetary panel of Rittenhouse's orrery in Philadelphia (figure 1). The models of the Rittenhouse orrery are probably the most complex 3D models ever made for such mechanisms.¹

My purpose, however, has never been to conduct research in 3D. I have been using 3D experimentally, and have developed my own tools and techniques. I have not used any commercial software; rather, I have used a set of tools which, when combined, make it possible to produce professional-quality 3D models. The most important of these tools has been mathematics. I have been computing shapes in 2D or 3D, I have not let software, or software libraries, do the computations for me.

During the past few years, I have also witnessed something new, namely a shift of attention of curators towards 3D. Heritage curators, who were often only marginally involved in the technical details, have become interested in the pedagogical use of 3D. At the same time, some clockmakers have teamed up with CAD enthusiasts and have created a number of models of clocks, and sometimes of other instruments. This has reached a point where the making of a 3D model often hides more important priorities, such as inventory and conservation. Making a 3D model sometimes even seems to be viewed as a substitute for proper documentation. One might say that 3D may hinder research, perhaps even block it.

For me, this is a serious danger, and this situation has led me to give more thought to the making of a 3D model, to its purpose, and to its sustainability.

2 The naïve approach to 3D dynamic models

Let me first consider the most common way of creating a 3D model. CAD software has made it relatively easy to create 3D models, and this has become very attractive to young horologists; it is even taught in some schools. 3D models are made both for existing clocks or mechanisms and as development tools for designing new mechanisms or clocks.

When making a 3D model of an existing clock, the clock is typically disassembled, its parts are measured, and each part is modelled in 3D.

¹Unless stated otherwise, all the 3D models I made were made privately and are not part of any university project.

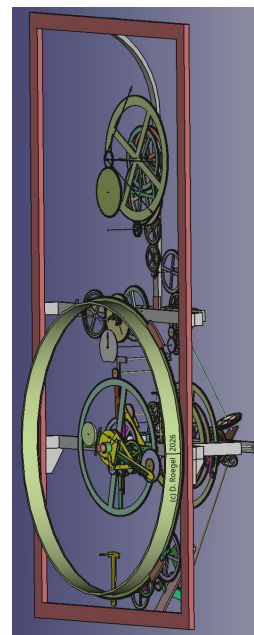
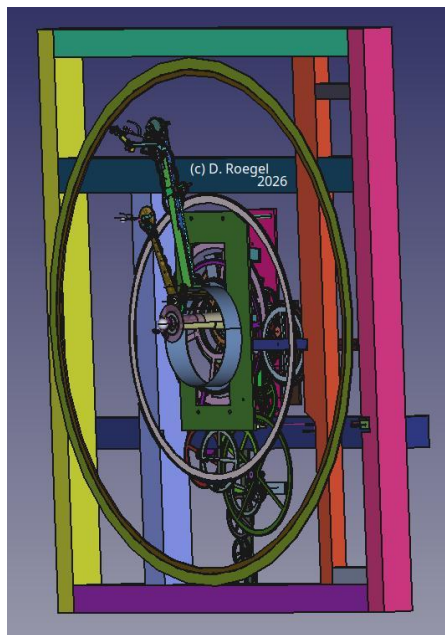
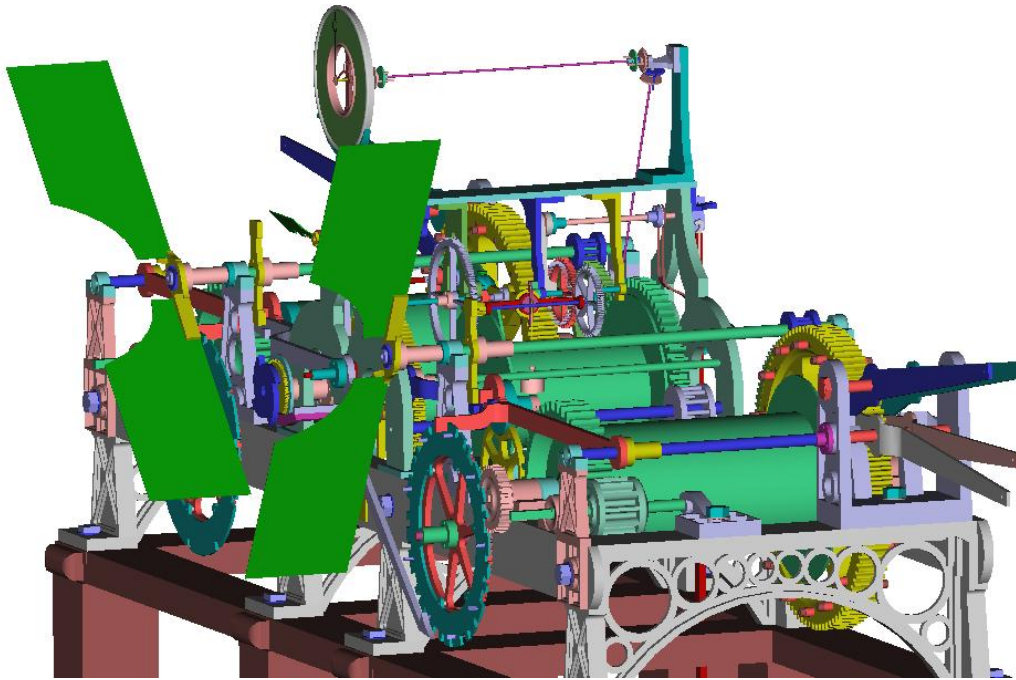


Figure 1: Top: the 3D model of the Paris Notre-Dame clock (2020). Bottom: the two panels of Rittenhouse's Philadelphia orrery (2026).

Sometimes, dimensions are taken directly from the assembled mechanism. The parts are then modelled independently, and the whole mechanism can be reassembled by aligning features belonging to different parts. CAD software packages often provide libraries of gear shapes, making their construction straightforward. Even when no such library is used, the shape of a tooth can be drawn in 2D and then replicated a number of times; the resulting 2D contour can subsequently be extruded to obtain the 3D shape of a wheel. Some parts are more complex to design than others, but in general, there are not many difficulties.

The models are usually mere transcriptions of the actual dimensions, and parameters are mostly unused. For instance, if a square plate with a side length of 10 cm is created and a hole is drilled at its center, the hole may simply be placed at position (5 cm, 5 cm). But if the dimensions of the plate are increased, the hole may no longer remain at its center. What is required in this case is the ability to distinguish between a hole positioned 5 cm from the edges of the plate and a hole positioned at the center of the plate. The hole could, however, be positioned according to other constraints: its distance from the edges might have to be equal to some other dimension in the construction, for instance, and dependencies may therefore have to be established.

In general, organizing such dependencies involves naming dimensions. This is seldom done, at least not systematically, because it is difficult and time-consuming. And even when it is done, it is often specific to a particular type of CAD software. As a consequence, most 3D models are rigid and lack flexibility and abstraction.

In any case, once the model has been created, it can be used to produce animations, or interactive versions can be made available through viewers, for instance using the glTF (Graphics Library Transmission Format) format. Some CAD software packages also provide the ability to animate the assembly of a mechanism simply by using the hierarchical structure of the assembly.

A number of 3D models have been produced, mostly by the Chronospédia project,² but also by the MIH (Musée International d'Horlogerie), by individuals such as Michel Dumain,³ and by a few others. Some of these models are available online.⁴

²<https://chronospedia.com>

³Most of Dumain's 3D models have been made in association with the *Association d'horlogerie comtoise*.

⁴When I write that a model is online, I usually mean that the model can be viewed through some sort of viewer. The actual source files of a model are in fact seldom available online.

In a few isolated cases, for instance at the Paul-Dupuy Museum in Toulouse, some clocks have been modelled with Blender. However, Blender is not a conventional CAD system and lacks some of the capabilities needed to develop models at a high level of abstraction.

The models that have been created in this way are usually used as pedagogical tools, sometimes through specially designed movies. However, not only do these movies fail to reflect all the features of the original mechanisms, they also usually mark the end of the project. The models are seldom developed further, and their authors move on to other projects. The models are also seldom accessible, and little is known about how they were created.

3 Issues for a 3D model

My own experience in 3D modelling, the analysis of what others have been making in 3D, as well as my work on the study and documentation of scientific heritage have led me to identify a number of issues which I consider important in the context of 3D models. Surprisingly, only very few of these issues are usually raised, and recent discussions mainly focused on vague notions of openness, conservation, and metadata. But there are other important issues to consider when planning to make a 3D model. These include, but are not limited to, the type of model; whether the model is parametric; its neutrality and replicability; its documentation, including documentation of its creation; the metadata associated with the model; how the model will be maintained; its accuracy and completeness; the production of animations; whether the actual mechanism is documented; whether there are plans for publication; and, more generally, the future of the 3D model.

3.1 The main issues

Let us now examine all these issues in more detail.

Documentation of the mechanism Even before a 3D model is made, it is essential to document the original mechanism. Of course, photographs are usually taken when the mechanism is disassembled, but, as far as possible, the documentation of the mechanism should be independent of the 3D model. I have examined a thousand turret clocks, taking measurements and photographs, making drawings, and never making a 3D model, because this was not my purpose. I

have also collected historical archives and other information relating to the mechanisms. Of course, such a survey can never be complete: there may be inaccessible areas (for instance, screw holes), some parts may be difficult to measure (for instance, screw threads), and it may also be difficult to determine how some parts fit together. Nevertheless, documenting a mechanism is an integral part of the overall project of modelling it in 3D.

If the mechanism is disassembled, it should in fact be documented at every stage: before disassembly (and, ideally, in each of its different locations), during disassembly, during restoration, if any, and afterwards. This documentation should, of course, be made available to everyone without restrictions. However, it should always be prepared with the possibility in mind of either constructing a copy of the mechanism or creating a 3D model. It is therefore best to make plans of all the parts, and these plans need not be produced using CAD. Pencil and ruler may be sufficient.

Documentation of the 3D model Likewise, making a 3D model is really about constructing an object, or reproducing an existing one. Some of the tools are similar to those used in an actual construction, while others are different. The documentation of the 3D model should contain not only the details of the choices made, but also what was created, when and by whom, why particular decisions were made, the mathematical relationships involved, the computations performed, and so on.

For some complex mechanisms, the construction is an exploratory process in which the mechanism is discovered and understood only gradually. This is what I experienced with the Rittenhouse orrery. There are many features that I understood only during the construction process. I also sometimes made mistakes or forgot to construct some parts. At other times, I made computational errors or had to move parts to avoid collisions. Sometimes it took months to understand a problem and determine how to solve it. If all of this is not documented, it will be lost, because it is not contained in the final model and is likely to be forgotten.

This is, in fact, somewhat like the process by which mathematicians prove important results. They may make mistakes, pursue unproductive approaches, go back, and try again, and when they eventually find a solution, they often erase all traces of their failed attempts. This produces a clean result, but it does not reflect the actual journey that

led to the solution.

Historical accuracy When a mechanism is modelled in 3D, various difficulties may arise. For instance, some parts of the mechanism may be missing, while others may not be original. It is therefore important to decide what is to be modelled, which precise historical state one is attempting to represent, and how missing parts should be modelled, if one decides to include them. Historical research and technical investigation may be necessary, especially since the person creating the 3D model may not be a historian or may not have experience working with such mechanisms.

It therefore seems preferable to bring together a number of people with different areas of expertise: experts on the mechanisms themselves, CAD specialists, historians, and so on. This is often difficult to achieve, since very few people may be familiar with particular mechanisms. Nevertheless, such collaboration is important for avoiding potential pitfalls. As an illustration, the (vanished) fourteenth-century clock of the *Palais de la Cité* in Paris has recently been modelled in 3D. However, the author of the model did not implement some of the features described in the plate used as a source for the model, and did not notice that many of the features included in the model do not belong to the fourteenth century. More importantly, the model presents itself as a representation of the first public clock in Paris, whereas this is still uncertain. These errors could have been avoided through more thorough historical research, greater experience with ancient turret clocks, and collaboration with other researchers.

Openness A model is not merely what one sees on a screen; it is not merely an animation. It is much more than that. A model is essentially what its creator manipulates, namely one or more source files. These source files contain information about the structure of the objects within the model, as well as the geometric features of each object. They contain information about the position of the objects in space, and they may also contain information about constraints, for instance linking one dimension to another. Providing a model only through an animation, or even through an interactive viewer, usually hides a great deal of essential information and is therefore not genuinely open. Whenever information is hidden, the user cannot do everything they might want or need to do with the model.

Some models are made available in exchange formats, such as STEP files, but even STEP files do not contain everything that should be

known about a model. They merely make it possible to transfer some of its geometry from one CAD software package to another.⁵ Therefore, it is incorrect to assume that converting, for instance, a SolidWorks source file to a STEP file will preserve all the information contained in the original source file. In general, it will not, except in simple cases. Likewise, providing glTF files is not a perfect solution, since these files no longer contain the original geometries. Finally, providing proprietary source files, such as a SolidWorks source file, is also not ideal, because most people will not be able to open or use them. Therefore, neither proprietary source files nor STEP files nor glTF files are genuinely open, even though making them available provides considerably more information than showing only screenshots or animations.

It is understandable that some authors of 3D models do not want their models to be freely available, because they may feel that doing so would mean losing control over their work. They could, however, at least provide STEP files while retaining the original CAD source files. I am, however, not particularly concerned about making CAD source files available, at least in STEP format. When I put the STEP files for the Paris Notre-Dame clock online in 2020, I expected that many people would be interested in them, but it turned out that, judging from the feedback I received, they were used by only two or three people. None of them seems to have built upon my work. It is possible that these models will be used by AI in the future, but so will many other things that we are producing.

Volumic/surface modelling A mechanism is made of a number of simpler parts that are assembled together. Each part is modelled separately and corresponds to a volume in space. This volume may be constructed in different ways. For instance, a cube can be created by assembling its six faces. However, when doing so, one must be very careful to produce a closed shell, rather than merely a shell that appears to be closed. When a shell is not actually closed, it has neither an interior nor an exterior. Once the shell has been created, it can be used to define an actual volume. There is a distinction between the shell and the volume it defines, since CAD software packages distinguish between surfaces and volumes.

⁵The STEP AP 242 format does provide the ability to store some constraints, and even kinematic information, but it does not preserve all of the design intent of the original model.

Another approach is to obtain volumes by combining other volumes or by applying specific operations to them. For instance, a cube could be obtained by extruding a square. Alternatively, it could be obtained from another volume by means of Boolean operations such as union, intersection, or difference.

Whatever the approach, it is essential to obtain actual volumes in the end, so that they can be processed further, in particular for 3D printing. A shell alone is not printable.

It should also be noted that some CAD software packages are more geared towards animation or rendering and less towards the creation of actual 3D models. It is therefore important to use the most flexible tool available to create the model itself, rather than starting with an animation.

Rigid/non rigid objects Most parts of a mechanism are rigid solids. For instance, a clock has a frame, gears, levers, screws, a pendulum, and so on. All of these are rigid parts. They may move, but they are not flexible. Only a few parts are non-rigid. In a clock, these include the suspension spring and a few other springs, such as those used for the ratchet pawls. In general, the deformation of such flexible parts cannot easily be represented in a STEP file and requires a higher-level representation. Of course, it is possible to model several distinct states of these flexible parts and interpolate between them when displaying the model.

Parametric models As mentioned above, the features of some objects in a model may depend on the features of other objects. Ideally, changing one feature should immediately affect the objects that depend on it. When the dependencies are simple—for instance, when two dimensions must be equal, or when one part of an object should be aligned with a part of another object—this is usually not too difficult to express, and CAD software may provide this functionality interactively. However, in more complex cases, the relationship between one feature and other features may require complex calculations that cannot simply be selected from a finite list in a drop-down menu. Many CAD software packages make it possible to include scripts for such computations, but when modelling an existing mechanism, it is often much simpler to ignore these constraints altogether. Very often, a person who creates a model of a machine, such as an old clock, is primarily interested in reproducing the existing mechanism rather

than in creating dozens of variants. Nevertheless, if the exact relationships between features are ignored, alignment problems may arise, partly as a result of measurement inaccuracies. Even when modelling a single mechanism, it is therefore preferable to parameterize it as much as possible—that is, to compute all dimensions that can be derived from others.

Part of the hindrance to the use of parameters lies in the dimensions themselves. If the dimensions were not given—if we were given only the radius of a wheel, R , its number of teeth, N , and so on—we would be forced to use parameters. We would have to compute the dimensions of other wheels from these parameters rather than from explicit numerical values. Although this approach is more difficult than the explicit, parameterless approach, it is also much more flexible and sustainable. In this sense, we could say that dimensions are the enemies of abstraction. I refer to the use of parameters and geometric computations as the algebraic-geometric approach.

Modularity When designing a model of a complex system, it is essential to adopt a modular approach. The idea is to define independent modules, each with its own inputs and outputs. This is somewhat analogous to object-oriented programming. For instance, in a clock with three trains, there might be three modules: the going train, which interacts with the quarter-striking train every 15 minutes, the quarter-striking train, and the hour-striking train, which is triggered by the quarter-striking train every hour. However, the external “interface” of a module consists not merely of the information it receives, but also of its physical shape, so that the module can be properly aligned with other modules. When a mechanism is complex, its modules may not be clearly distinguishable, or they may have numerous spatial interactions. This is typically the case in an orrery, where it is difficult to consider the trains associated with each planet entirely separately, especially since they often share a number of parts. Nevertheless, whenever possible, a complex mechanism should be divided into smaller parts, each of which is simple enough to be handled by a single person. In a clock, each train is sufficiently simple that it does not need to be divided further, and an entire clock could therefore be regarded as a single module.

But even when a mechanism is treated as a single module, it still contains smaller parts, each of which is essentially independent. A given wheel can easily be constructed independently of another, except that

some of its parameters may be related to parameters of other parts. For instance, the primitive radius of one wheel is usually computed from the primitive radius of another wheel. Similarly, the size of the central hole of a wheel is related to the size of the arbor passing through it. If these features are properly specified, different parts can be built independently, and they can even be built by different people. Some of the clocks modelled as part of the Chronospédia project were, for instance, modelled by different students, each of whom modelled different parts in SolidWorks. It would even be possible to take this idea further, with different parts being built by different people using different CAD software. There is in fact no need for the same software to be used, as long as the interface between the objects is well designed.

CAD software packages usually provide the ability to design a model hierarchically, as a tree of components. The top level represents the entire mechanism, with each subsequent level adding further detail. For instance, a clock may be divided into three parts: the going work and the two striking works. The hour-striking work may then be divided into the frame and the other parts, or perhaps into the frame and the arbors. Each arbor can in turn be divided into its components. This is a very convenient approach, as it makes it easier, in particular, to isolate individual parts of a mechanism. However, it does not make the interactions between different parts any easier to manage. For instance, the hour-striking work is typically triggered by the quarter-striking work, which means that an internal part of the hour-striking work must be associated with an internal part of the quarter-striking work. Moreover, the hour-striking work is sometimes triggered a second time by the going work. Thus, we have not only relations between elements belonging to separate components or modules, but also spatial relations between them. Taking all these relations into account in a parametric way, within a modular and hierarchical structure, and with a view to creating an open system, is not a simple task.

Sustainability and maintenance 3D models are often presented as static objects. Someone creates a 3D model of a mechanism, and that is it. The model is supposed to be perfect, final, and not in need of further improvement. For instance, within the Chronospédia project, a number of models have been created using SolidWorks, and once a model has been created, it is no longer modified. Alas, almost every model I have seen has some shortcomings or errors. The models

I made of the Rittenhouse orrery in Philadelphia are no exception. When I made them, I had access only to photographs, and in a number of cases I was unsure how some of the parts were connected. With a better knowledge of the actual mechanism, it may become desirable to improve at least some details of the 3D model.

Thus, 3D models are not carved in marble; they are not perfect sculptures that no one should ever touch. What has been created is therefore not merely a 3D model, but a 3D model *with* shortcomings that may need to be addressed. How, then, can a model be improved? There are two main ways.

The simplest is to modify the model itself: to fix errors, replace some parts, adjust some dimensions, or perhaps add missing parts. For instance, in one case on the Chronospédia website, a clock is shown with its count wheel flipped.⁶ The wheel needs to be reversed. To address such a problem with greater certainty, however, the model should be made available so that anyone can correct it, rather than relying solely on its creator, who may be busy with other things.

Another way to improve a model is to construct an entirely new one—that is, to replicate it, perhaps reusing ideas from the original model. This may be the only solution when the original model is not available. However, it can only be done if the original model has been made available with sufficient detail.

Thus, when we speak of “sustainability,” what we really mean is that a model can remain usable over a sufficiently long period without having to be entirely reconstructed. This presupposes that the 3D model is accessible and open. The notion of “sustainability” is, in fact, similar to that which applies to domestic appliances. A 3D model is sustainable either when it is perfect—which is never the case—or when it can be improved. Otherwise, it is not sustainable. This means that any non-open 3D model is *de facto* not sustainable.

Neutrality An important issue is the ability to create neutral models—that is, free and open 3D models that do not require proprietary CAD software and that can be maintained by an open community. A 3D model should not be dependent on a particular CAD software package, as is the case with the Chronospédia project. The same is true of many other 3D models, such as those created by the MIH, which appear to be used only within the museum.

⁶See <https://www.chronospedia.com/s/chronospedia/item/28595>, but this link is probably not permanent.

Publications Although a 3D model may be created merely for one's pleasure or for the production of educational films in museums, it is unfortunate that many 3D models are never put to further use, not even as a basis for comprehensive descriptions of the mechanisms for which they were created. This is the case, for instance, with the partial model of the Strasbourg astronomical clock, made with Autodesk 3ds Max; the animations created with Blender for the Musée Paul-Dupuy in Toulouse; the models displayed on the Chronospédia project website; and many others.

To a great extent, this limited use results from the fact that those who create 3D models are not necessarily researchers and may have little experience with scholarly publication. Of course, there are exceptions. One example is the work of Michel Dumain, some of whose models have been used to describe the original clocks in the bulletin of the AFAHA, the French horological association. Another reason why models are not further exploited is that many of those who create them either do so professionally or seek to produce as many models as possible. Publishing a description of a mechanism, particularly a clock and especially without a historical background, can also be difficult and, of course, is rarely financially rewarding.

Nevertheless, despite these sometimes understandable reasons, I believe that publication should be kept in mind, especially when a 3D model is created. The absence of a publication describing a mechanism represents a loss for research, just as the absence of adequate documentation for almost any restoration represents a loss. A 3D model can make it much easier to explain how a mechanism works, and it can make it possible to show configurations that cannot be shown with the actual mechanism. This opportunity should not be missed. This is, for instance, what I intend to do with the Rittenhouse orrery in Philadelphia.

Accuracy It is important to decide how accurate a model should be, or rather, how faithful it should be to the original mechanism. There is usually a trade-off between faithfulness and complexity. The more details a model incorporates, the more complex and time-consuming it becomes to create and maintain. There are also many details that may be inaccessible, as well as dimensions that may be unknown, and one is therefore often led to simplify the model. In my case, when I made the model of the Paris Notre-Dame clock, I tried to be as faithful as possible, using my photographs and the dimensions I had taken

from the clock. There are still discrepancies, and not everything has been modelled, but the model looks very much like the original clock. For the Rittenhouse orrery, I had a number of photographs, but the mechanism is much more complex, and I also had to make choices about what to model and how. In some instances, the photographs did not show all the details, resulting in a few shortcomings, although these do not affect the general description of the orrery.

In any case, the choices made should be clearly understood and documented, and they should be such that the resulting shortcomings can be addressed at a later time. For instance, someone may want to add some screws in the future or change the shape of some part of the Rittenhouse orrery, and the model should make this possible.

Animations Once a 3D model has been created, if it is well made, it can be animated. Nowadays, some CAD software can animate a mechanism by taking its contacts into account. When one wheel is rotated, the wheels with which it meshes automatically rotate as well. This is very convenient and makes it easy to create animations. Moreover, many CAD software packages can animate the assembly of a 3D model, so producing a movie is no longer as difficult as it once was.

On the other hand, simply letting the CAD software perform the animation is not particularly gratifying. Where, then, is the knowledge of the creator? In my case, I have made several animations featuring the Paris Notre-Dame clock, both as standalone videos on YouTube and as an Android application, and I have *not* let the CAD software create the animations for me. In my animations, I have computed the motion of every moving part, as well as the motion of the camera. Finally, the videos I made were actually produced by assembling several thousand still images and adding a soundtrack. No CAD software made the videos for me!

Completeness When a model is made available, it should provide more than access to the shapes of its parts or the ability to visualize them; it should also provide the necessary technical information about each part. This is especially important for gears. For each gear, it should be possible to determine the number of teeth, the radius, and other properties, such as the position of the part in space. It is important that these attributes can be obtained easily, for instance by selecting a part in an assembly. This is what I implemented in my Android application in 2021, and also in the glTF file that I made available online in 2025. Although this is not difficult to implement, I know

of no other viewer that provides the same features, at least as of September 2026.

Future of a 3D model We have to be realistic: once a 3D model has been created, it is unlikely that anyone will continue to develop it, and almost every 3D model eventually becomes a dead end. From what I have observed in the Chronospédia project, for instance, errors are rarely corrected. Moreover, correcting errors in a 3D model means not only correcting the model itself, but also updating the related animations and other forms of documentation.

As far as my own models are concerned, as time passes, I forget the details of their construction, and it becomes increasingly difficult for me to delve back into their intricacies. This makes it unlikely that either I or others will make substantial changes to my models. An enthusiast might perhaps rebuild a model from scratch, but modifying a complex model is difficult and risky, since new errors may be introduced, and is therefore not necessarily advisable.

What we can aim for, however, is to develop 3D models in such a way that they can be corrected, improved, and analyzed with the help of AI. For instance, my models are based on many explicit computations, whereas other models are usually based on implicit computations. I believe that AI could help me reorganize these computations, make them more understandable, and perhaps improve the models themselves. In the case of other, more static models, AI could be used to identify their underlying mathematical properties and make them more abstract. I do not think, however, that a human would easily be able to do this alone.

Thus, if a model is well organized, I expect AI to be able to improve its organization, detect problems, add missing parts, analyze its behavior, and even produce documentation or animations. AI could therefore assist in the maintenance of 3D models, provided that the models are structured in a way that makes this possible.

We have now considered the most important issues related to 3D models. Unfortunately, as I have noted, many of these issues are neglected or ignored.

3.2 Which CAD should be used?

In this note, I am not going to advocate for any specific CAD software. In fact, the choice of CAD software is less important than the openness of the

model, the ease with which it can be parameterized, the ease with which its entire construction process can be documented and made available, its sustainability, and so on. Some CAD software packages may make these things easier than others, but with most decent CAD software, it should be possible to achieve the goals described above.

3.3 About the “FAIR” principles

In the above list of important issues concerning 3D models, I have not mentioned the so-called “FAIR” principles for open science.⁷ According to Wikipedia,

The FAIR principles emphasise machine-actionability — the capacity of computational systems to find, access, interoperate, and reuse data with none or minimal human intervention

These principles can also apply to 3D models stored in a repository. They primarily concern the way in which such models are described, accessed, and made available for reuse, rather than the intrinsic qualities of the models themselves. FAIR stands for Findable, Accessible, Interoperable, and Reusable. If a model follows these principles, “Findable” means, for instance, that the model is deposited in an archive and accompanied by rich metadata. “Accessible” means that the model should be openly accessible and that it should remain possible to access it in the long term. And so on.

These are all rather obvious principles, but I noticed, at a presentation of the Chronospédia project at a university library in December 2025, that the audience was very interested in these principles, much more so than in the models themselves. This highlights another danger of 3D models, somewhat paralleling the situation with digitized archives: people tend to look at them in a very shallow way, and library professionals may be more interested in Dublin Core metadata than in the openness, accuracy, documentation, or sustainability of the 3D models, or even in the preservation of scientific heritage, as if all these things were already guaranteed.

4 The path towards openness and reusability

Let me summarize the main issues I have discussed by presenting the different stages in the development of a 3D model as a progression towards openness and reusability. Again, I must stress that the more open a model

⁷See https://en.wikipedia.org/wiki/FAIR_data

becomes, the less control its creator retains over it, and the more third parties can learn about how the model was developed. This is why models are often not made entirely available, even by people who advocate “open science”: making a model fully available means relinquishing some control over it. Consequently, some creators will intentionally choose not to go beyond certain stages in the progression I sketch below.

In what follows, some of the stages may overlap, and there is sometimes no strict order between them.

1. The first stage is usually the creation of a closed model using some CAD software, such that no one else has access to the model itself. Perhaps some screenshots can be made available. This is currently the stage of the model of the Rittenhouse orrery in Philadelphia, although I intend to make it more open.
2. A second stage is to produce non-interactive animations, for instance to explain how a mechanism works. Sometimes nothing more is produced by the creator of the 3D model. However, beautifully rendered animations are not sufficient. They can be beautiful and impressive, but they are almost always incomplete and represent a dead end. The animations I have recently seen in museums and elsewhere, for instance those for the Strasbourg astronomical clock or those made for the Paul-Dupuy Museum in Toulouse (the Habrecht and Janvier clocks), convey only general ideas and are not sufficient to understand the complete construction or all the details of these clocks. Someone who wants to study these clocks needs more.
3. A third stage is to provide an interactive 3D model, for instance in glTF format. This is the case with the models that have been made available online by the Chronospédia project since 2024.
4. An easy improvement over the third stage is to provide an interactive model in which individual parts can be queried for metadata, such as the number of teeth of a gear. This is what I implemented in January 2025 in the 3D model I made of the Paris Notre-Dame clock.⁸ As of September 2026, this feature has not been implemented in the Chronospédia models.
5. A further improvement, which can occur in addition to an earlier stage, is to provide extensive documentation of the models, as well as of the original mechanism.

⁸<https://horloges.github.io/3D/notredame.html>

6. A further improvement is to provide the model in an exchange format, such as STEP. This is what I did in 2020 with the Paris Notre-Dame clock.⁹ As of September 2026, the Chronospédia project still does not provide any STEP files.
7. A small but important improvement over the previous stage is to provide split STEP files, that is, one STEP file per part, in such a way that the files can be reassembled. This is also what I did in 2020 with the Paris Notre-Dame clock, and it does not seem to have been done by anyone else.¹⁰
8. Further stages include the printing of a 3D model (which I did for the Paris Notre-Dame clock in 2021), the creation of animations incorporating the motion of the internal parts (also done in 2021 for the Paris Notre-Dame clock¹¹), and the development of mobile applications with interactive animations, interactive metadata queries, and dynamic simulation (all done for Android in 2021 for the Paris Notre-Dame clock¹²).
9. A further stage towards openness is to provide the complete source files, for instance SolidWorks source files, Blender scripts, or equivalent files for whatever software was used. This has seldom been done, and it was done neither by me nor by the Chronospédia team, which has used both SolidWorks and Blender.
10. But it is still possible to go further, because a 3D model does not contain the entire construction process of an object. Separate documentation of the 3D model, its entire chronology, the computations involved, and so on would make it possible for others to reconstruct the model, reproducing every step of its construction, in the same or another software. Others could then even improve the model.
11. At this stage, it should be possible to use AI to go further. AI should make it possible to correct the model, explain it, and develop it beyond what the human creator could reasonably achieve.

⁹<https://github.com/roegeld/notredame>

¹⁰It should be noted that, at the beginning of 2025, I gave clear indications to the Chronospédia team on how this can be achieved with SolidWorks.

¹¹<https://www.youtube.com/@horlogenotredame/videos>

¹²<https://roegeld.github.io/apk/notre-dame-clock-0.9.apk>. The APK file used to be on the Google PlayStore, but was removed since I didn't update it sufficiently often.

5 Conclusion

Making a 3D model for a complex mechanism goes far beyond providing the CAD files, ensuring their preservation, or following the “FAIR” principles. A 3D model should be put into context: it is only a small part of a complex landscape. A 3D model should not be improvised; its future development should be kept in mind. It should be open, useful, and reusable—that is, it should provide a basis for further work, rather than merely being accessible.

A 3D model should not be thought of as a replacement for traditional documentation, but as a supplement to it. And in the case of scientific heritage, a 3D model should not divert attention from the real priorities of scientific heritage, which are inventory by researchers, documentation, and conservation.

A 3D model should also provide easy access to the mechanism for everyone, rather than merely being available to a few people or to people who have access to a particular type of software. It should be possible to navigate easily within the model, and this navigation should not be restricted to a static animation, as is often the case.

The openness of a 3D model means that the construction information should be available; it means that the technical details of the model should be available, and that the choices, computations, and so on should all be available and freely reproducible by anyone. In other words, once a 3D model has been made, a great deal of work still lies ahead!